higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 2

(12030152)

12 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 5 pages.

NCV

TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Keep subsections of questions together.
5. Rule off across the page on completion of each question.
6. Clearly cross out ALL work you do NOT want to be marked.
7. Write neatly and legibly.

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QUESTION 1: BASIC CONCEPTS AND ROAD-BUILDING TERMINOLOGY

Define each of the following terms:

1.1 Formation level

1.2 Asphalt

1.3 Road signs

1.4 Verge

1.5 Road shoulder

(5 × 2) **[10]**

QUESTION 2: ROAD SURVEYING

2.1 Define the term *line level*. (3)

2.2 Name TWO tools that can be used to level the surface of a road. (2)

2.3 Briefly explain FIVE points to remember when using a line level. (5)

2.4 What does the abbreviations BC and EC stand for? (2)

2.5 Explain in detail how to set out a curve using a small radius. (4)

2.6 A dumpy level is one of the machines used to level the ground.

Explain in FOUR points how to set out a dumpy level. (4)

[20]

QUESTION 3: SOIL GRAVEL AND ROAD-BUILDING MATERIALS

3.1 Differentiate between *moisture content* and *optimum moisture content*. (2 + 2) (4)

3.2 Explain in detail how to test a soil-plasticity index. (6)

3.3 List THREE things needed when performing a cone-penetrometer test on soil. (3)

3.4 Give TWO reasons why water is needed during soil compaction. (2)

3.5 What causes soil to swell? (2)

3.6 State THREE consequences of soil swell. (3)

[20]

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QUESTION 4: ROAD DRAINAGE

4.1 Give TWO purposes of drainage facilities on the road surface. (2)

4.2 List THREE areas where a trench drain is used. (3)

4.3 Write down the function of each of the following drains:

4.3.1 Culvert

4.3.2 Mitre drain

4.3.3 Stone pitching

(3 × 2) (6)

4.4 What dangers can soil erosion on the road cause? (2)

4.5 Give THREE main sources of water. (3)

4.6 Name TWO types of drainage parallel to the road surface and describe the purpose of each. (4)

[20]

QUESTION 5: EARTHWORKS

5.1 Name TWO plants for each of the following earthwork categories and give the uses of each:

5.1.1 Excavation

5.1.2 Earthmoving

5.1.3 Compaction

(3 × 4) (12)

5.2 Most hazards are created by human errors or machine malfunctions.

State THREE things that must be done to avoid a tragedy during earthwork. (3)

5.3 Which precautionary measures can be taken on site to prevent each of the following accidents:

5.3.1 Falling

5.3.2 Head injuries

5.3.3 Falling materials

(3 × 1) (3)

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5.4 Discuss the practical measures to mitigate hazards in the workplace under each of the following headings:

5.4.1 Excavation

5.4.2 Safety programmes

5.4.3 Equipment operation

(3 × 2) (6)

5.5 Explain the term *earthwork*. (1)

[25]

QUESTION 6: LAYER WORKS

Briefly discuss FIVE steps to consider when applying a stabiliser to a layer of soil. **[5]**

TOTAL: 100

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